

CY 559: Machine Learning in Chemistry

Course Code: CY 559

Course Name: Machine Learning in Chemistry

L-T-P-C: 2-0-2-3

Prerequisites: No prior experience in artificial intelligence and programming is required—just curiosity, passion, and a willingness to experiment with data and molecules.

Intended for: BS 3rd/4th Year, MSc, PhD

Mutual Exclusion: None

Distribution: Elective (Odd/Even)

Approval: 61st BoA

Preamble

The course Machine Learning for Chemistry introduces the fundamental concepts and methodologies of machine learning with a strong focus on practical applications in chemical sciences. Rather than emphasizing only theoretical aspects, the course adopts a hands-on, example-driven approach designed for students eager to apply machine learning techniques to real chemical problems.

It is aimed at a broad audience, from theoretical chemists interested in exploring data-driven approaches, to experimental chemists looking to integrate machine learning into their research workflows.

The course begins with a concise overview of the core principles of machine learning, followed by a discussion of how chemical systems can be represented computationally, including molecular representations and the role of symmetry. Students will then explore key learning paradigms (supervised and unsupervised learning) and major model architectures, ranging from linear regression and kernel methods to artificial neural networks.

Throughout the course, theoretical insights will be anchored in practical examples drawn from modern chemical research, such as materials discovery, catalyst design, spectroscopy, and microscopy image analysis, demonstrating how machine learning is reshaping the way we study and innovate in chemistry.

Course Modules

- **Module 1: Foundations:** What is Machine Learning? Types of ML (supervised, unsupervised), Basic ML workflow (data → model → validation → deployment), Data in chemistry: sources, structure, and challenges, Basic Python data handling: NumPy, pandas, matplotlib, Introduction to chemical data representation (SMILES, InChI, molecular graphs) (7 hours)

Practical: Handling small chemical datasets (e.g., solubility, boiling points), Exploratory Data Analysis (EDA) using pandas and matplotlib.

- **Module 2: Data Representation in Chemistry:** SMILES and Molecular Graphs, Molecular descriptors and fingerprints (MACCS, Morgan), RDKit and descriptor generation, Feature scaling, dimensionality reduction (PCA, t-SNE), Feature selection and correlation analysis. (7 hours)

Practical: Generating and visualizing molecular fingerprints, Applying PCA on chemical property datasets

- **Module 3: Unsupervised Learning:** Clustering: k-Means, hierarchical clustering, DBSCAN, Dimensionality reduction revisited (PCA, t-SNE, UMAP), Outlier detection in experimental datasets, Chemical space visualization and diversity analysis. (6 hours)

Chemical Applications: Clustering of similar compounds in drug/molecule libraries, Reaction space analysis and catalyst clustering

Practical: Visualizing chemical space using PCA/t-SNE, Cluster analysis of organic molecules

- **Module 4: Supervised Learning:** Linear regression, Polynomial regression. Regularization (Ridge, Lasso), overfitting, logistic regression, k-NN, decision trees, random forest, and SVM. Model training (Cross-validation, train-test split, hyperparameter tuning) and evaluation metrics (MAE, RMSE, R^2 , accuracy, ROC, confusion matrix). (9 hours)

Chemical Applications: Predicting molecular solubility, QSAR (Quantitative Structure–Activity Relationships), Spectral data classification (IR/NMR peak prediction)

Practical: Building regression models to predict physical properties (e.g., solubility, boiling points), Model performance comparison using scikit-learn.

- **Module 5: Neural Networks and Deep Learning:** Basics of Artificial Neural Networks (ANNs), Activation functions, loss functions, optimization, Introduction to TensorFlow/PyTorch, Deep learning for chemistry: property prediction, spectra, and reaction yield prediction, Concept of graph neural networks (GNNs) and message-passing networks. (6 hours)

Chemical Applications: Predicting molecular energy from SMILES or graphs, Spectra prediction (IR/Raman), Reaction outcome prediction (introductory level).

Practical: Building a simple feedforward neural network using TensorFlow/Keras.

- **Module 6: Case Studies and Advanced Topics:** ML in materials discovery and catalysis, ML in computational chemistry (potential energy surfaces, molecular dynamics), ML for spectroscopy and image analysis (microscopy, IR maps), Interpretability of ML models (SHAP, LIME), Data ethics, reproducibility, and bias in chemical ML. (7 hours)

Practical: Mini project: students choose a dataset (e.g., QM9, MoleculeNet), Train, validate, and interpret a simple ML model

Laboratory/Practical/Tutorial Modules

Practical components are included in the respective modules.

Text Books

- Introduction to Machine Learning with Python* – A. Müller & S. Guido
- Machine Learning in Chemistry: The Impact of Artificial Intelligence* – Hugh M. Cartwright
- Machine Learning in Chemistry* – Jon Paul Janet and Heather J. Kulik, American Chemical Society, 2020
- Machine Learning in Chemistry: Data-Driven Algorithms, Learning Systems, and Predictions* – Edward O. Pyzer-Knapp and Teodoro Laino, American Chemical Society, 2019.

References

Online Resources: MoleculeNet datasets, DeepChem tutorials, RDKit documentation, MIT's "ML for Materials" course notes

Data Sets: MolGPT: <https://github.com/devalab/molgpt>

MolDis: <https://moldis.tifrh.res.in/>